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Jared Paul

Data Scientist

Portfolio
github.com/jrpaul08
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Education

McMaster University Hamilton, ON
B.Eng., Software Engineering, *summa cum laude*

Skills

Languages: Python, R, SQL, Bash, Java
Technologies: pandas, scikit-learn, SciPy, PyTorch, PySpark, Matplotlib, Seaborn, Power BI, MongoDB, MySQL, Flask, React

Experience

Preteckt | Data Scientist 09/2023 – 02/2025

- Performed exploratory data analysis on time-series telematics data using SQL queries, Python, pandas, and NumPy to uncover relationships between sensor streams, failure patterns, and data anomalies.
- Partnered with in-house technicians, the system's end users and key stakeholders, to scope customer complaints, using root cause analysis on sensor data to build a threshold-based predictive maintenance alerting system.
- Ran threshold experiments that backtested alert triggers on historical data and shadow-tested them against production, achieving over 80% precision across 300+ daily alerts.
- Developed a scikit-learn preprocessing pipeline incorporating imputation, outlier filtering, feature scaling, and PCA for GMM-based sensor anomaly detection.

McMaster University | Research Assistant 05/2023 – 09/2024

- Replicated results for a mixed-integer optimization model for nurse scheduling using the INRC-II benchmark, accounting for staffing coverage, shift assignments, nurse preferences, and workload restrictions across 4- and 8-week planning horizons.
- Evaluated the model on 360 weekly scheduling problems, finding provably optimal solutions in 274 cases (76%) and achieving an average 1.90% optimality gap on the remaining 86.

Projects

READING BUDDY — AI READING COMPANION | *Python, PyTorch, Hugging Face, RAG, FastAPI, Modal* 06/2026 – Present

- Developed an end-to-end speech-to-speech generative AI pipeline integrating Whisper, Qwen2.5-14B, and Kokoro to deliver spoken, natural-language responses in about 5 seconds.
- Built a spoiler-safe RAG system pairing unstructured book text with structured character and event data generated via prompt-engineered AI agents (Claude Code), grounding LLM responses without exposing future chapters.
- Deployed independently scalable GPU-backed microservices on Modal, leveraging parallel container warmup and persistent model caching to achieve low-latency at about $0.04 marginal compute cost per question.

MULTI-LABEL CHEST X-RAY CLASSIFICATION | *Python, PyTorch, MLflow, scikit-learn, Docker, FastAPI, Azure* 09/2024 – 01/2026

- Built a PyTorch-based multi-label classification pipeline to detect 13 thoracic diseases from the NIH ChestX-ray14 dataset.
- Trained and optimized a MobileNetV2 convolutional neural network with hyperparameter tuning and dynamic thresholding, achieving 0.80 ROC-AUC and 60% recall on multi-label disease detection.
- Integrated MLflow for experiment tracking, logging hyperparameters, evaluation metrics, and model artifacts to enable reproducible training and systematic experiment comparison.
- Developed and deployed a containerized FastAPI inference service using Docker and Azure Container Apps to serve real-time chest X-ray predictions via a public API endpoint.

NBA ORACLE | *Python, JavaScript, pandas, scikit-learn, Playwright, BeautifulSoup4, Flask* 06/2025 – 09/2025

- Developed an end-to-end machine learning pipeline to predict NBA game outcomes from over 3,400 games across three seasons.
- Implemented a web scraping system using BeautifulSoup and Playwright to asynchronously collect and structure team and player statistics into a time-series dataset.
- Engineered statistical features, including rolling averages of team performances and temporal aggregates capturing team form, offensive/defensive efficiency, and recent trends.
- Trained a Ridge Classifier with sequential feature selection and time-series cross-validation, achieving 71% prediction accuracy.

RETAIL CUSTOMER SEGMENTATION — RFM ANALYSIS | *Python, TypeScript, pandas, scikit-learn, Matplotlib* 08/2024

- Performed RFM analysis & EDA to derive Recency, Frequency, and Monetary features and uncover actionable customer segments.
- Detected and separated outliers using the IQR method to preserve high-value customers while maintaining clustering quality.
- Implemented K-Means clustering with feature scaling, using the Elbow method (inertia) and Silhouette score to determine the optimal number of clusters, resulting in four primary segments plus three specialized outlier clusters.
- Discovered that about 25% of customers generate majority revenue and 35% fall in the "Nurture" segment with potential to grow.